

Sophia LaSala

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EDUCATION

Farmingdale State College

B.S., Electrical Engineering Technology

- **GPA: 3.7**; Dean's List, Member of IEEE Student Chapter

Expected: May 2028

Farmingdale, NY

WORK EXPERIENCE

Undergraduate Research Assistant

Farmingdale, NY | July 2026 – Present

- Conduct undergraduate research on the design and characterization of low-cost MEMS microphone arrays as alternatives to laboratory-grade measurement microphones.
- Develop software to automate frequency response measurements and experimental data collection.
- Design and assemble electronic hardware for multiple microphone array configurations.

Empire State Service Corp.

Farmingdale, NY | Aug. 2026 – Present

- Conduct monthly civil discussions on current issues, encouraging respectful dialogue among students.
- Support student affairs initiatives and coordinate programming for the campus community.

Information Technology Intern

North Massapequa, NY | Jan. 2024 – May 2024

- Supported deployment and configuration of Chromebooks and iPads, including device resets and basic troubleshooting.
- Assisted with routine IT operations and device management in a professional environment.

ASSET Conference Student Presenter

Melville, NY | March 2024

- Presented a student-led session on 3D modeling and LiDAR scanning technologies to educators and school board members.

PROJECTS

Raspberry Pi Portable MP3 Player – *Raspberry Pi, Python, CAD, 3D Printing*

Aug. 2026 - Present

- Developing a portable MP3 player using a Raspberry Pi and Python, with support for locally stored music and metadata.
- Designing and 3D modeling a custom enclosure while integrating the display, controls, and electronic components with consideration for power consumption and portability.

Graph-Based Indoor Navigation System – *C#, Dijkstra's Algorithm*

Dec. 2025 - Jan. 2026

- Developed an indoor navigation system in C# to determine routes between locations within a mapped environment.
- Implemented Dijkstra's shortest-path algorithm to calculate routes through interconnected locations.

Traffic Light Control System – *Arduino, C++, Embedded Systems, Electronics*

Nov. 2024 - Dec. 2024

- Built an embedded traffic light control system using an Arduino, LEDs, LCD display, and buzzer, implementing timed state transitions to simulate real-world traffic signal operation.

LEADERSHIP & ACTIVITIES

Last Hope Animal Shelter Volunteer - *Wantagh, NY*

May 2022 - Present

Plainedge eSports Club President - *North Massapequa, NY*

Sept. 2023 - June 2024

Plainedge FIRST Robotics Awards Coordinator - *North Massapequa, NY*

Sept. 2023 - June 2024

SKILLS

- **Programming:** C++, C#, Java, Python, SQL
- **Tools/Hardware:** Soldering, Breadboarding, Circuit Design, CAD, Arduino IDE, Git/GitHub